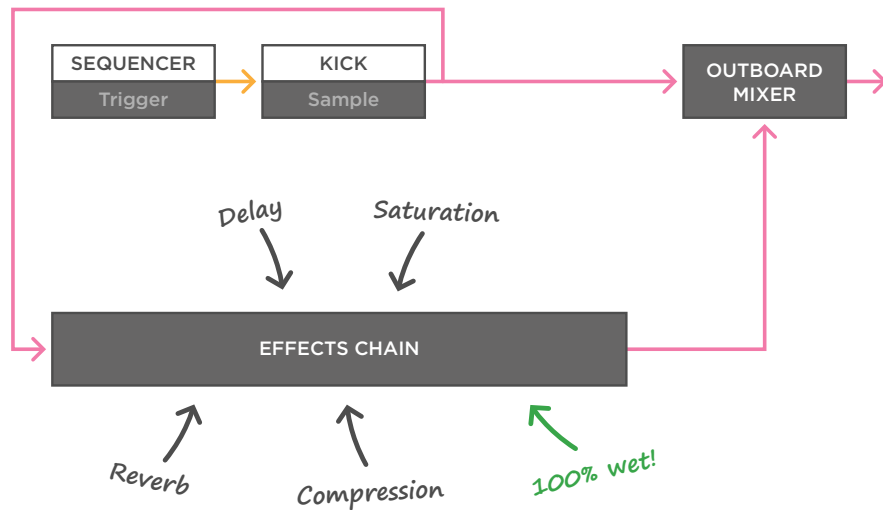


Techno Rumble

Simple Rumble

Audio
Control Voltage
Trigger / Gate

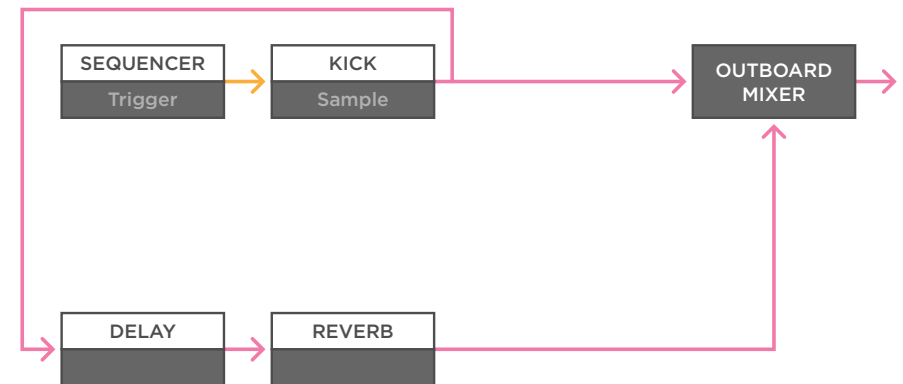


(a)

a - This is the basic setup. A sequencer is used to trigger a single kick. Send the kick directly to an outboard mixer, so you always have access to the clean kick in your mix. Then, a copy of the kick is send through a series of effects and processors, to create a cloud of sound related to the kick.

You can experiment with pretty much any kind of effects you like. However It's important to alter the signal drastically, so there is no phase cancelation with the dry kick. More dramatic effects that have a 100% wet output are preferable. Send the result to the outboard mixer, so you can mix it in with the clean kick to taste.

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(b)

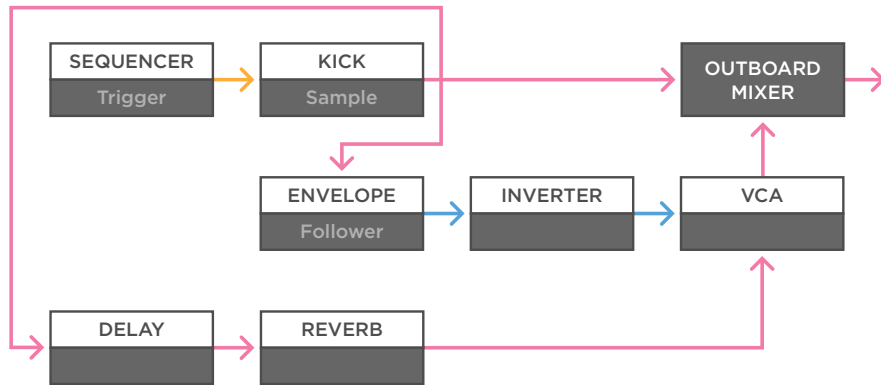
b - It works well to start your chain with a full wet time based effect like a delay. This way you can create rumble based on a groove that is timed around the kick. When you mix these signals together again to make your track, they won't fight each other to much.

You can experiment by adding or swapping effects in the rumble chain. A clean delay gives a pretty percussive result. Run the full wet delay through a full wet reverb if you like more smeared sounds.

Techno Rumble

Rumble Sidechaining

Audio
Control Voltage
Trigger / Gate

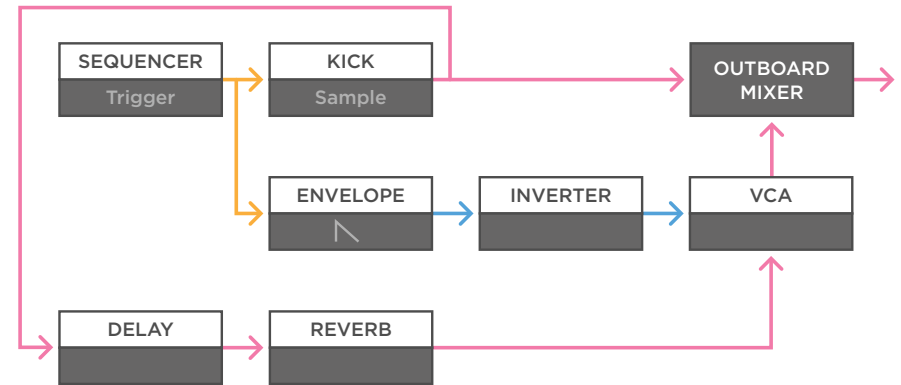


(a)

a - In order to keep the kick clean and powerful in the mix, and create a pumping groove. You need to duck the sound of the rumble every time the kick hits. To do so, add a VCA at the end of the rumble chain.

Then there are two common ways to create side chaining. The first is to make another multiplication of the audio of the kick, run that through an envelope follower, invert the envelope, and use it to close the VCA and lower the sound of the rumble, whenever the kick is audible.

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(b)

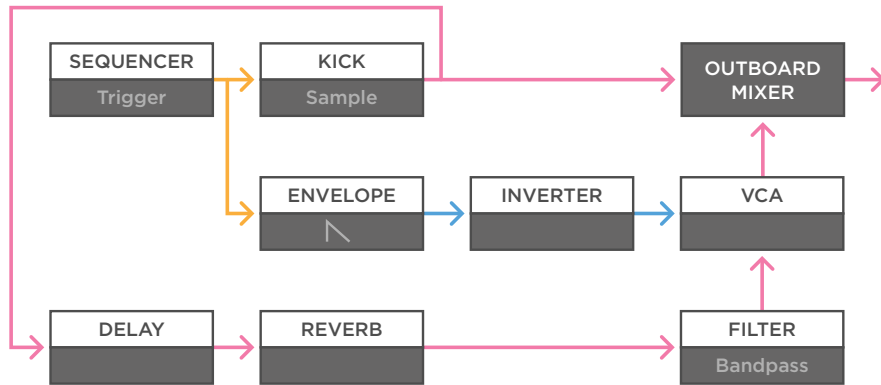
b - To get more control over the attack and release time of the ducking, you can also make a copy of the trigger to the kick, and use that to trigger an attack decay envelope. Again, invert it, and send it to the VCA to duck the rumble. You can use the attack and decay time of the envelope to create a more pumping sound.

Because there is a VCA at the end of the chain to keep the kick clean, you can experiment with higher feedback settings to sculpt denser sounds.

Techno Rumble

Advanced Rumble

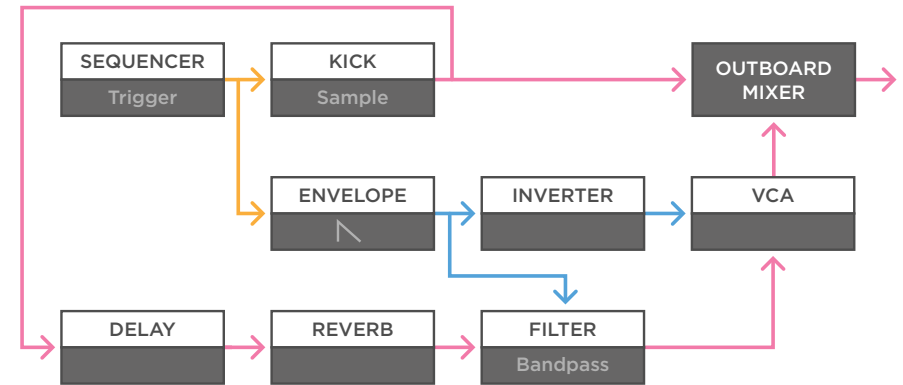
Audio
Control Voltage
Trigger / Gate



(a)

a - To get more control over the sound, you can add a separate filter to the chain, right before the VCA. For lower rumbles a low pass filter will do the job. But a bandpass filter is even more flexible. The more control over resonance or feedback, level or drive the filter offers, the more ways have to sculpt the sound. So experiment with different filters.

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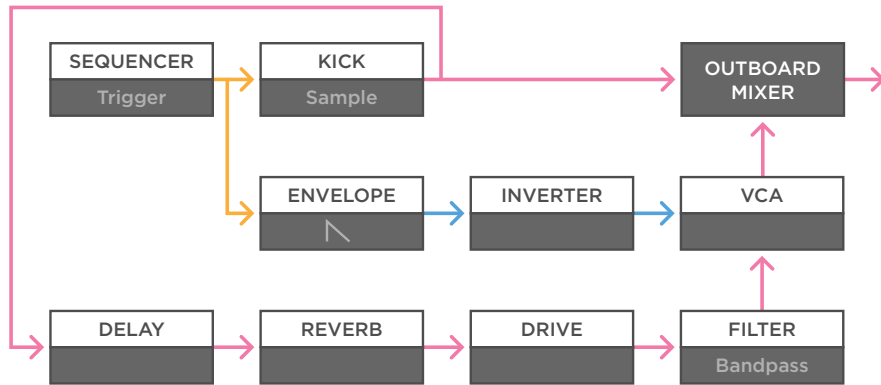
(b)

b - An easy mistake is to leave out the VCA, and use an inverted envelope to close a low pass filter to create a side chaining effect. This will work in some way, and may have an audible nice effect. But when a low pas filter is used, the volume difference mostly comes from filtering the mid frequencies, while you especially want to lower the volume of the low frequencies of the rumble when the kick hits. However, with a bandpass filter you can experiment by sending a copy of the envelope to the filter to cut out the low end, and create interesting movement in the sound.

Techno Rumble

Advanced Rumble

— Audio
— Control Voltage
— Trigger / Gate

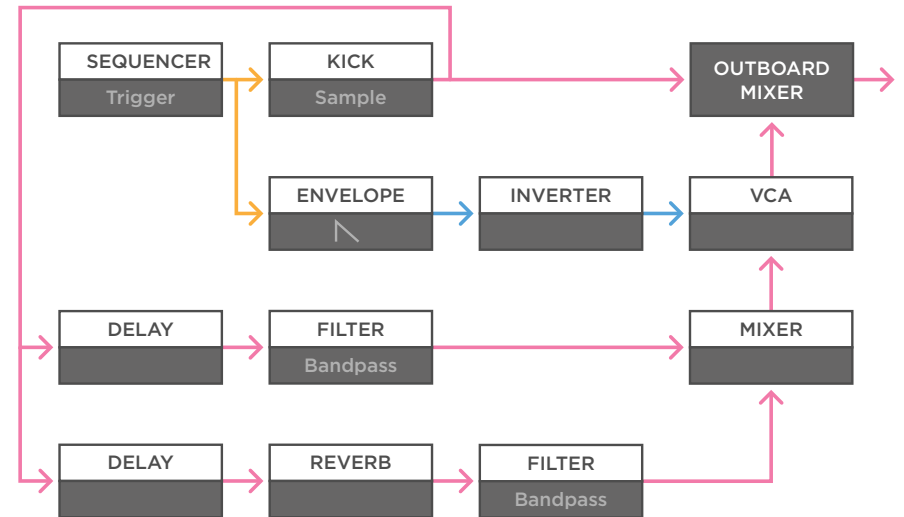


(a)

a - Another effective way to shape the rumble is by using saturation or distortion. Some filters might offer enough drive, but a good distortion can really add a lot of grid to the sound. You can experiment by placing the distortion before the VCA, as it is shown here, or after the VCA.

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TECH TALK

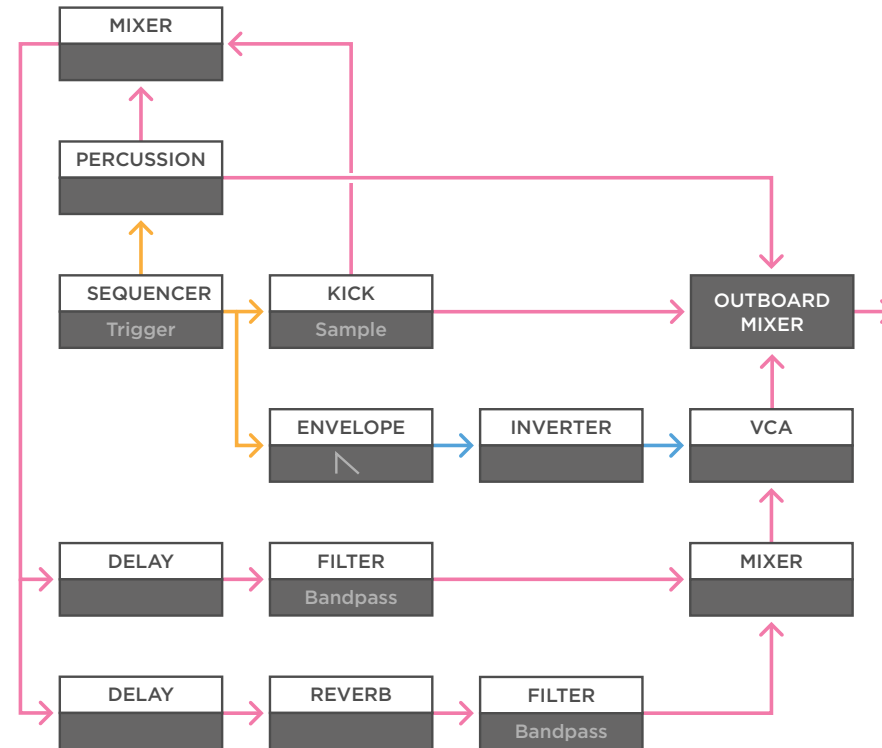


(b)

b - Instead of using one rumble chain you can send a copy of your kick into multiple chains. For example, use one chain with a delay and bandpass filter for a more percussive rhythmic sound. Then, use a delay, heavy reverb with some pitch shifting, and a bandpass filter for some air in your rumble.

Send the different chains into a modular mixer, and send the mix into a VCA for side chaining. You can use the mixer to sculpt the rumble over the course of a track or performance.

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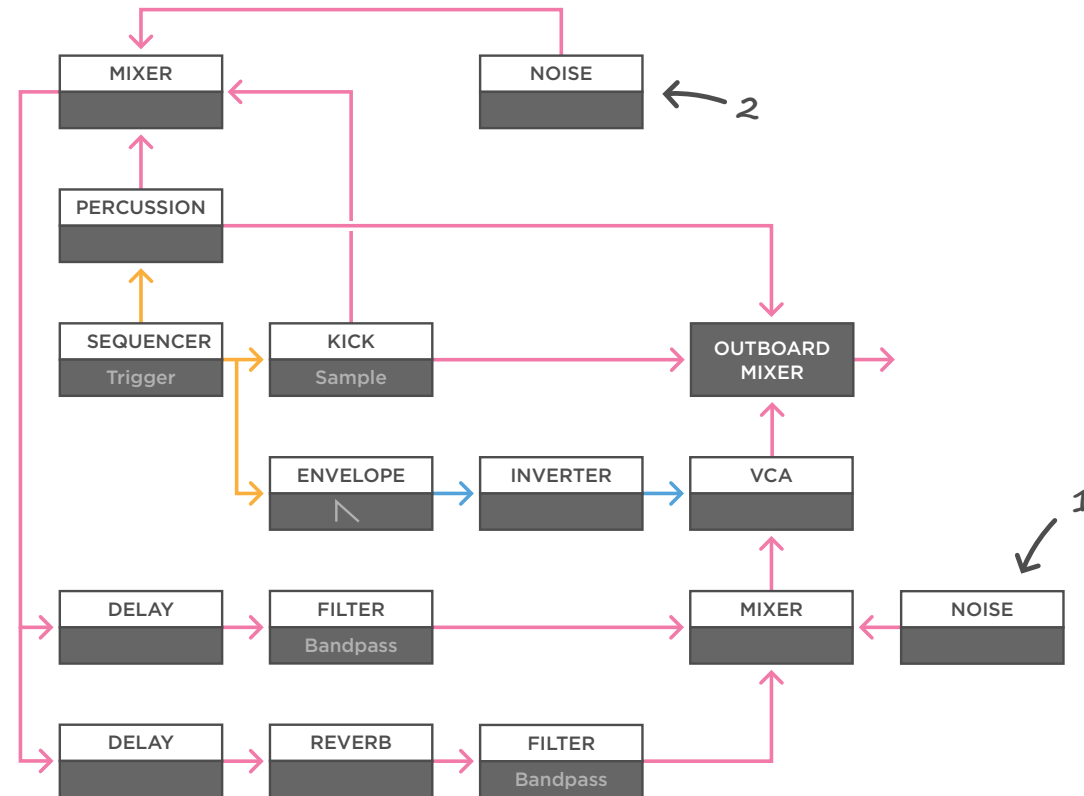
a - You can also use a mixer before the rumble chain to mix multiple sounds into your chain. For example, add a percussive sound to the setup, and just like with the kick, send a clean version into your outboard mixer. Then, send the kick and percussion, or of course any sounds you like, to the mixer before the rumble chain. You can use that mix as the source of the rumble chain, and manually tweak it during a performance or recording to sculpt the sound.

Techno Rumble

Advanced Rumble

MONOTRAIL TECH TALK

Audio
Control Voltage
Trigger / Gate



(a)

a - Here is a larger patch example, with a mixer both before and after the chain. You can use samples, or noise to add grit to the rumble. You can use slow, but still audio rate sample and hold signals. They can add lovely gritty dark textures. But you can also use white noise for a more open sound.

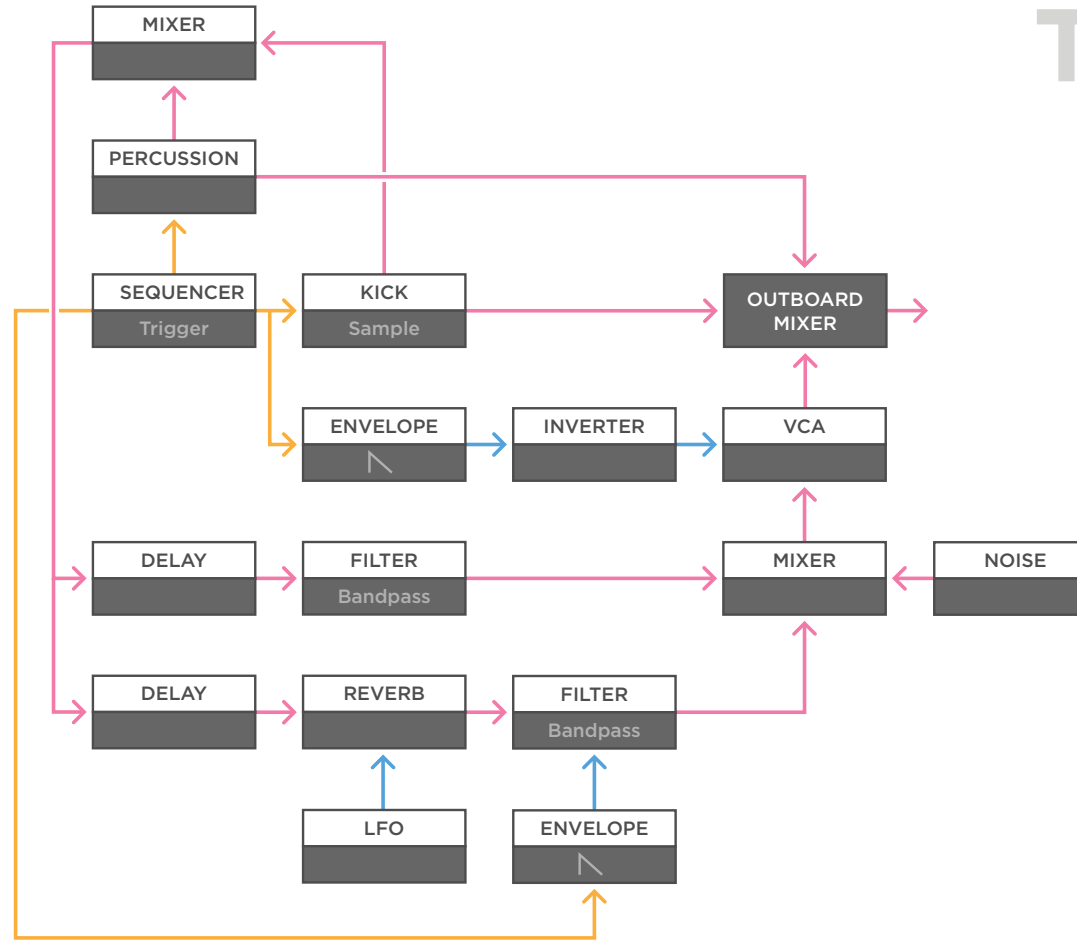
Experiment with mixing in the noise before the VCA at the end of the rumble chain, keeping it textured and fairly clean (1). Or mix it in at the front of the rumble chain. In this case of course the noise is also processed by all the effects, and smeared out (2).

Techno Rumble

Advanced Rumble

MONOTRAIL TECH TALK

Audio
Control Voltage
Trigger / Gate



(a)

a - And then you have a rather extensive patch that holds a lot of elements that can be modulated. There are many ways to add motion to this setup, here are just two examples. To add some simple movement you can use an LFO to any audible parameter. For example the drive or gain on a reverb. And it's also worth it to experiment with time based modulation. For example, take a copy of the clock, use it to trigger an envelope, and use that to modulate a bandpass filter.